

# Introduction to Computer Simulation II

## Homework 08

Due: Tuesday, 09 June 2026

### 15. Least-squares fits

The  $M$  parameters  $a_1, a_2, \dots, a_M$  of a least-squares fit  $y(x_i) = y(x_i; a_1, \dots, a_M)$  to  $N$  data points  $(x_i, y_i)$  with statistical errors  $\sigma_i \equiv \epsilon_i$  are determined by minimisation of the quantity

$$\chi^2 = \sum_{i=1}^N \left( \frac{y_i - y(x_i; a_1, \dots, a_M)}{\sigma_i} \right)^2,$$

the so-called “chi-square”. For a fit to a straight line,  $y(x_i) = y(x_i; a, b) = a + bx_i$ , the two parameters  $a$  and  $b$  can still be determined explicitly. Reproduce this relatively easy analytical calculation.

The following data are taken from recent research work jointly with Marco Müller and Prof. Desmond A. Johnston (Edinburgh) on the so-called “gonihedric” Ising model (Nucl. Phys. B **888** (2014) 214), for which theoretical arguments predict for large lattice sizes  $L$  a dependency  $\beta_{\max}(L) = a + b/L^2$ . Test this prediction with the help of least-squares fits (sometimes also called “chi-square” fits) to the  $N$  data points using fit ranges from  $L_{\min} = 8, 10, \dots$  to  $L_{\max} = 24$ . The fit ansatz is acceptable when  $\chi^2/\text{dof} \approx 1$ , where dof denotes the degrees of freedom, that is in this case  $N - 2$ .

| $L$ | $\beta_{\max}$ | $\epsilon$ |
|-----|----------------|------------|
| 08  | 0.51818        | 0.00010    |
| 10  | 0.52932        | 0.00010    |
| 12  | 0.53562        | 0.00011    |
| 14  | 0.53967        | 0.00010    |
| 16  | 0.54232        | 0.00008    |
| 18  | 0.54421        | 0.00009    |
| 20  | 0.54533        | 0.00008    |
| 22  | 0.54649        | 0.00005    |
| 24  | 0.54726        | 0.00007    |

#### 16. Fits of the susceptibility for the 2D Ising model

Use the single-cluster algorithm to simulate the susceptibility of the 2D Ising model (using the “improved estimator”) for a  $128 \times 128$  square lattice in the high-temperature phase from about  $0.99\beta_c$  to  $0.85\beta_c$  in steps of  $0.01\beta_c$  (this can be easily extended to higher temperatures, e.g., to  $0.70\beta_c$ ).

Using this data try to estimate the critical exponent  $\gamma$  from least-squares fits with the ansatz

$$\chi(\beta) = a(1 - \beta/\beta_c)^{-\gamma},$$

assuming the known exact value of the transition point  $\beta_c = \ln(1 + \sqrt{2})/2$ .